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LCSeg-Net: A low-contrast images semantic segmentation model with structural and frequency spectrum information

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ABSTRACT

Semantic segmentation in low-contrast images is a challenging problem due to the ambiguous boundary of the segmented target and indistinguishable noise. Current models generally segment the target with local features, which leads to the lack of structural information, which reduces the performance. And they directly fuse multi-scale features to keep details, while it may integrate noise into the fused features. These problems reduce the performance of segmentation in the low-contrast image. To solve the above issues, we propose an image semantic segmentation model called Low-Contrast Segmentation Net (LCSeg-Net). The model enhances the structural information with the global context and reduces the noise of the fused features through the adaptive fusion way. Meanwhile, to improve the robustness of LCSeg-Net, we augment the low-frequency spectrum of the fused feature in the training phase. Extensive experiments are conducted on the five public datasets. Results show the proposed model achieves the best comprehensive performance in the low-contrast image.

1. Introduction

Image Semantic Segmentation (ISS) aims to classify each pixel in the image [1], which divides the image into multiple regions. It plays a very significant role in image analysis and has practical applications in many fields, such as remote sensing [2], intelligent logistics [3], autonomous vehicles [4], and video surveillance [5], etc. Due to its importance, ISS has been widely studied, which results in a lot of interesting achievements. However, as a special research direction of ISS, semantic segmentation in low-contrast images, such as the medical image [6], surface defect [7], and smoke segmentation [8], is still a challenging problem.

Currently, most ISS models utilize the Convolution Neural Network (CNN) and the skip connection to segment the target. Through accumulating the convolution layer with pooling and activation functions, the CNN-based models effectively extract and obtain the rich semantic feature from the input images for accurate segmentation [9]. Additionally, these models generally use the skip connection to retain detailed features [10]. These features from shallow layers are fused with the features from deep layers, which reduces the information loss caused by the down-sample process in the model. However, these models still have two problems, which reduce the accuracy of segmentation in low-contrast images. (1) In the low-contrast image, the boundary of the segmented target is indistinct. Meanwhile, the convolution filter only extracts the local features, resulting in the lack of structural information in the CNN-based models. This makes it difficult for the ISS models to accurately predict the target region. (2) The texture of the noise of low-contrast images and the segmented target is similar. Since the features of shallow layers are directly fused in the skip connection, it makes the fused features contain a lot of noise [11]. It also reduces the accuracy of the segmentation result. As shown in Fig. 1(a), the pixels of the smoke are similar to that of the background, resulting in the ambiguous boundary of the smoke. The ambiguity leads to the inaccurate prediction of U-Net, as shown in Fig. 1(d). Moreover, as illustrated in Fig. 2(a) and Fig. 2(c), the noise in the red box is ambiguous, which induces the U-Net segment the noise as the target. The above challenges make traditional ISS models hard to deal with low-contrast images.

To solve the problems mentioned above, we put forward a Low Contrast Segmentation Net for low-contrast image semantic segmentation, or LCSeg-Net for short. To mitigate the insufficient structural information, LCSeg-Net utilizes the global context to enhance the CNN-based feature. Meanwhile, to reduce the impact of the noise, the model adaptively guides the feature fusion in the skip connection, which suppresses the noise of the fused feature. In the model, the SIE block is designed to extract the global contexts with multiple Multi-Layer Perceptron (MLP) layers, which enhances the features with the structural information. In previous research, scholars find that Transformer-based [12] and MLP-based [13] models learn global contexts, which emphasizes the

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(a) (b) (c) (d) (e)

Fig. 1. Illustration of the samples of the smoke image and the corresponding ground truth, boundary, and segmentation result. (a) The smoke image. (b) The ground truth of the smoke image. (c) The boundary of the smoke. (d)The segmentation result from U-Net. (e) The segmentation result from LCSeg-Net. As shown in these figures, LCSeg-Net achieves more accurate segmentation result than U-Net does with the enhanced structural information.

(a) (b) (c) (d)

Fig. 2. Illustration of the samples of the polyp image and the corresponding ground truth, and segmentation result. (a) The polyp image. (b) The ground truth of the polyp image. (c) The segmentation result from U-Net. (d) The segmentation result from LCSeg-Net. As shown in these figures, LCSeg-Net achieves more accurate segmentation result than U-Net does by suppressing the noise.

structural information of the input image. These models improve the performance of the dense prediction tasks with the global context, such as ex-ViT [14], MLP-Mixer [13], and GasHis-Transformer [15]. However, the self-attention mechanism and the MLP structure of these models are quadratic computational complexity, which limits practical applications in real-world scenarios. To avoid the high costs as these models, we apply the MLPs along horizontal and vertical dimensions to jointly obtain the global contexts. These MLPs initially select the spatial location of the features, which should be computed for interaction. It provides the global context by the adaptive selection along the spatial dimension, which enhances the structural information of the features. Note that the process only needs linear computational complexity. Meanwhile, we utilize the boundary of the target to make the SIE block obtain accurate structural information. As shown in Fig. 1(e), LCSeg-Net accurately segments the smoke with enhanced structural features. To reduce the noise in the skip connection, we design the FSF block to fuse the features from the encoder and the SIE block in the frequency domain. It utilizes the parameterized matrix to suppress the noise of fused features. Moreover, as shown in Fig. 2(a), the noise is mainly similar to the target on the color and texture, which are dependent on the low-frequency components of the features. Therefore, to make the proposed model robust to complex changes of noise, we model a distribution for the low-frequency components of fused features. After that, we resample from the distribution to generate the new low-frequency spectrum. The new low-frequency spectrum is combined with the high-frequency spectrum to generate the new fused features, which improves the robustness of the block for the complex noise.

We summarize our main contributions as follows: (1) We propose LCSeg-Net to accurately segment the targets in the low-contrast image. The model utilizes the global context to enhance structural information, which accurately captures the boundary information of the low-contrast target. To distinguish the noise and target, the model reduces the noise of the fused feature in the skip connection. (2) We design the SIE block to enhance the structural information of CNN-based features in LCSeg-Net. In this block, multiple MLPs are utilized to efficiently enhance the structural information with global contexts, which only needs linear computational complexity. Furthermore, the block utilizes the boundary information of the segmented target to improve the accuracy of the structural information. (3) In LCSeg-Net, the FSF block is designed to reduce the noise in the skip connection. It filters the noise of the fused features by the parameterized matrix. Additionally, to improve the robustness, we model the distribution for the low-frequency spectrum of the fused features. The distribution is utilized to resample the new low-frequency spectrum, which makes LCSeg-Net adaptive to the texture change of the noise. (4) We evaluate the performance of the LCSeg-Net for the low-contrast image segmentation across five public datasets, including the polyp, the nuclear, the skin lesion, the surface defects of magnetic tile, and the smoke. The results show the LCSeg-Net achieves the best Accuracy, Dice, and IoU score compared to other State-Of-The-Art (SOTA) models and while keep practical efficiency, which proves the LCSeg-Net accurately segments the targets of the low-contrast images.

2. Related works

With the successful application of deep learning in various fields of computer vision, a lot of models based on deep learning have made amazing achievements in ISS. Generally, these studies mainly focus on how to utilize the different contexts of the image and preserve the detailed information of the image to obtain accurate segmentation results.

For the first type of the studies, a naive method is to extract the spatial context by enlarging the Effective Receptive Field (ERF) of the model [16]. Since the features extracted by the convolution limit the spatial context, to obtain more spatial context, current CNN-based models generally utilize the method that increase the depth of ISS models or using modified CNN to enlarge the ERF. For example, Yuan et al. used the ResNet [17] as the backbone to expand the size of the ERF, which makes the models achieve competitive performances on five public datasets [18]. However, due to the increment of parameters, the method reduces the efficiency of the model. To efficiently enlarge the ERF, Chen et al. constructed Deeplab family models based on the atrous convolution. Initially, they only used the atrous convolution to extract features, which enlarges the ERF with the same parameters as the convolution [19]. After that, they utilized the multi-scale features with the atrous spatial pyramid pooling to better segment the multi-scale objects [20]. To further refine the segmentation results, Chen et al. proposed Deeplabv3+. The model increases the efficiency of the

model with the depth-wise separable convolution [21]. Even though the above methods have modified the size of the ERF, the property of the convolution filter still limits the ERF of these models. Inspired by studies in natural language processing, some ISS models take advantage of the self-attention mechanism to obtain the global ERF [22]. For example, Wang et al. proposed the Non-local net that utilizes the global ERF to improve the accuracy of the downstream task [23]. The model computes the relation between all pixels to obtain the global context with the non-local block. Unfortunately, though these models improve the accuracy of the segmentation with the global ERF, the prohibitive computational cost hinders their usages in many real applications.

Additionally, some studies use other contextual information, such as boundary and object contexts to improve the performance of the ISS model. Li et al. proposed DecoupleSegNet to jointly learn the body and boundary of the segmented target [24]. In this model, the features are divided into the body feature and the boundary feature. The model improves the performance of segmentation by jointly optimizing these features. To improve the performance with the object context, Yuan proposed the Object Context Net (OCNet) to enhance the contribution of the same object, which reduces the impact of the irrelevant object [18]. The model utilizes two sparse matrixes to efficiently approximate object contexts, which pays attention to the necessary object information and reduces the interference of the redundant information. However, these studies ignore the impact of the noise in the low-contrast image.

For the other type of studies, they mainly focus on how to preserve the detailed features of the images. To keep detailed features as much as possible in the down-sample process of ISS models, a lot of interesting studies have been made. U-Net utilizes the skip connection to preserve the detailed feature, which effectively improves the accuracy of biomedical image segmentation [10]. To improve the performance on the basis of U-Net, Zhou et al. proposed a model named U-Net++ that aggregates the features of varying semantic scales with nested and dense skip connections to accurately segment multi-scale targets. Subsequently [25], U-Net3+ uses full-scale features and deep supervision to fuse features of different scales [26]. These methods preserve multi-scale details, and yet they still cannot capture enough detailed features as they fail at extracting the overcomplete features. To solve the problem, KiU-Net extracts the overcomplete features with the overcomplete convolutional structure, with which a good ISS performance is obtained in the 2D and 3D image datasets [27]. However, the overcomplete convolution structure of KiU-Net needs huge memory costs. As a result, the KiU-Net is difficult to apply to high-resolution images. Meanwhile, to decrease the redundant information in the skip connection, some researchers also integrate the context information into the skip connection. For instance, Yuan et al. proposed a Classification-assisted Gated Recurrent Network (CGRNet) to segment the smoke [28]. To capture accurate contextural information, the model utilizes multi-scale contextual information to enhance the feature in the skip connection. This makes it accurately predict the smoke of the image by capturing the multi-scale contextual information of the smoke. Differently from CGRNet, UTNet utilizes global contextual information to improve the representation of the feature in the skip connection with the modified multi-head self-attention block, which enhances the segmentation results with cheap computation costs [29]. Unfortunately, to save the computation cost, the model compresses some useful information, which reduces the accuracy of results. Different from the above studies, the proposed model efficiently utilizes the enhanced structural contexts and adaptively fused features to improve the accuracy of the segmentation.

3. LCSeg-Net

In this section, we present our method in four subsections. First, we illustrate the overall architecture and implementation details of the LCSeg-Net. Second, we elaborate on the SIE block, which enhances the structural information with the global context. Third, we introduce the FSF block that adaptively fuses the features in the frequency domain. The block suppresses the noise of the fused feature. Meanwhile, it resamples the low-frequency spectrum of the fused features, which improves the robustness of the model. Last, we introduce the loss function of the proposed model, which supervises the model in the training phase.

3.1. Network architecture

To effectively implement ISS in the low-contrast image, we propose the LCSeg-Net that utilizes the structural information and low-frequency spectrum of the image to improve the segmentation performance. From Fig. 3 it can easily be noticed that the model is composed of the encoder, SIE block, and FSF block. The encoder is used to extract the spatial features of the image, which are recovered to the segmentation results. Due to the CNN-based encoder, the features from the encoder lack structural information. It decreases the ISS performance in low-contrast images. Therefore, we use the SIE block to enhance the structural information with the global context. Meanwhile, the block is supervised by the boundaries of the segmented target in the training phase, which improves the accuracy of the structural information. Furthermore, to retain the detailed features of the image during the process of encoding the image, LCSeg-Net fuses the features from the encoder and SIE block. However, it introduces noise into the fused features, which decreases the accuracy of ISS in low-contrast images. To reduce the noise in the skip connection, we utilize the FSF block to remove the noise of the fused features in the skip connection. Additionally, we randomly sample new features from the distribution of the fused features to improve the robustness of the model within the block.

3.2. Structural information enhancement block

The ISS models generally extract the feature with the convolution filter. However, the convolution filter only can extract the local features. It emphasizes texture information rather than structural information. However, in the low-contrast image, due to the indistinct boundary of the target, structural information is important. Therefore, we employ the SIE block to enhance the structural information with the global context.

As shown in Fig. 4, the block consists of multiple MLP layers. These MLP layers are utilized to obtain the global contexts and the boundaries of the target. Meanwhile, these features are supervised by the boundary information to improve their representation. To extract the global context, we extract the features along the channel, horizontal, and vertical dimensions by Channel MLP, Horizontal MLP, and Vertical MLP respectively. In detail, the Channel MLP is utilized to compute along the channel dimension of the input features $X \in \mathbb{R}^{H\times W\times C}$ with height H, width W, and channel C as:

$$f_{channel}(X) = X \cdot W_c + b_{channel} \tag{1}$$

where $W_c \in \mathbb{R}^{C\times d}$ and $b_{channel} \in \mathbb{R}^{H\times W\times d}$ denotes the weight and bias of the Channel MLP respectively. It maps the dimension of feature X from $H \times W \times C$ to $H \times W \times d$ with the matrix multiplication.

For the spatial dimensions, directly computing the results as the formula (1) leads to huge computational costs with large resolution features. Therefore, to efficiently extract the global context on the spatial dimension, we initially feed X into the MLP layer to predict the C offsets of the location, which makes the model adaptively incorporate spatial information. The process is formulated as follows:

$$O_{vertical}(X) = X \cdot W^{offset}_{vertical} \tag{2}$$

$$O_{horizontal}(X) = X \cdot W^{offset}_{horizontal} \tag{3}$$

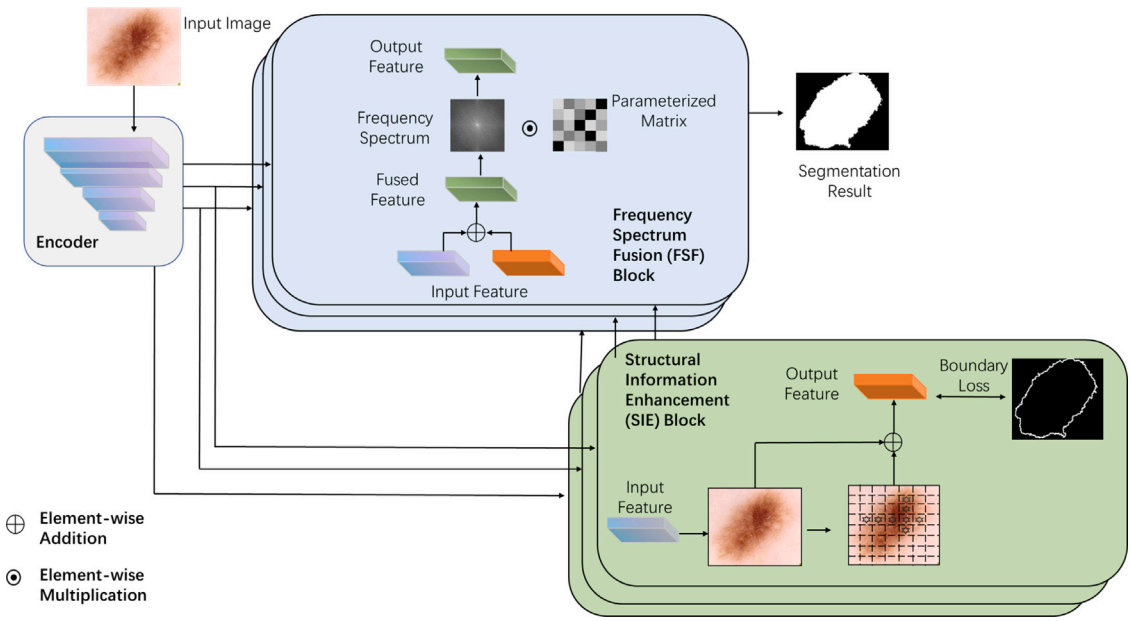

Fig. 3. Network overview of LCSeg-Net. The model utilizes the structural enhancement feature to improve the accuracy of the prediction. Meanwhile, to avoid the false prediction by the noise, the model suppresses the noise in the skip connection. The model consists of the SIE block and the FSF block. The SIE block enhances the structural information of the feature with the global context. To keep the accuracy of the structural information, the block is supervised in the training phase by the boundary of the target. The FSF block adaptively fuses the feature in the frequency domain with the parameterized matrix, which reduces the noise of the fused feature.

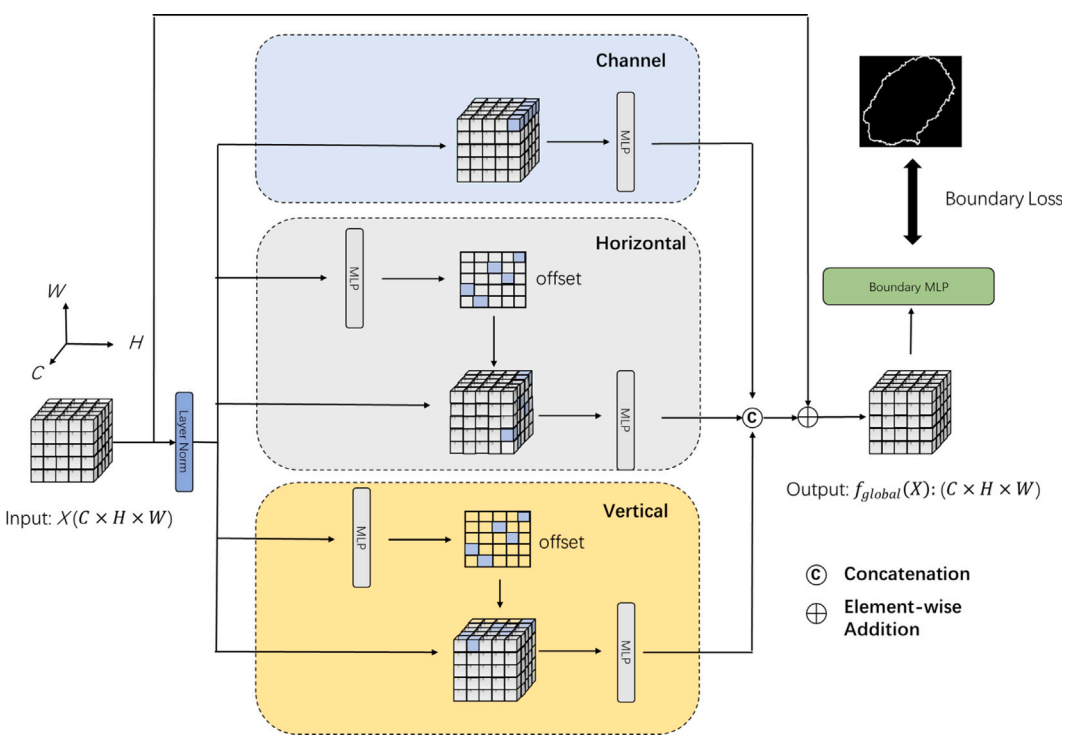

Fig. 4. Overview of the SIE block. The block enhances the structural information with the global context. The block fuses the features from the Channel MLP, Horizontal MLP, and Vertical MLP to obtain the global context. Especially, the Horizontal MLP and Vertical MLP predict the offsets to adaptively select the spatial locations of elements that are computed in the corresponding MLP. Furthermore, to improve the accuracy of the structural information, the block is supervised by the boundary of the target.

where $W^{offset}_{horizontal} \in \mathbb{R}^{C\times C}$ and $W^{offset}_{vertical} \in \mathbb{R}^{C\times C}$ denotes the weight of MLP layers that predict offset along width and height respectively. Due to the unlimited offset, the structural information with the global context is supplied to the model. We compute the spatial feature along the horizontal and vertical dimensions with the offset prediction. For example, for the vertical dimension, the element $X_{i,j,c}$ is the value of cth channel at the spatial position (i, j). We use the Vertical MLP to process $X_{i,j,c}$ as:

$$f_{vertical}\left(X_{i,j,c}\right) = \sum_{c=0}^{C} X_{i+O_{vertical}\left(X_{i,j,c}\right),j,c} \cdot W^{vertical}_{c,:} + b_{vertical} \tag{4}$$

The Vertical MLP computes the $X_{i,j,c}$ with the offset $O_{vertical}\left(X_{i,j,c}\right)$. $W^{vertical} \in \mathbb{R}^{C\times d}$ and $b^{vertical} \in \mathbb{R}^{H\times W\times d}$ denotes the weight and bias of the Vertical MLP. Likewise, the process of the Horizontal MLP is

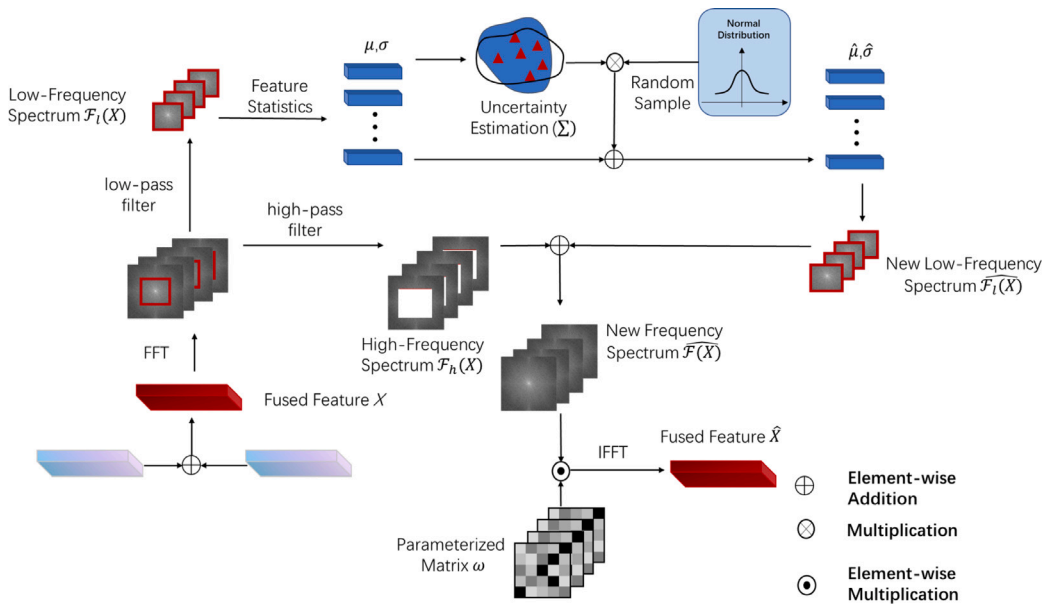

Fig. 5. Illustration of the FSF block. The block suppresses the noise of the fused features X. It initially converts the X to the frequency spectrum $\mathcal{F}(X)$ with the 2D Fast Fourier Transform (FFT). After that, the frequency spectrum is multiplied by the parameterized matrix ω, which adaptively emphasizes the useful information of the fused features. Meanwhile, to improve the robustness of the model, we utilize the low-frequency component $\mathcal{F}_l(X)$ of the $\mathcal{F}(X)$ to generate the new frequency spectrum $\widehat{\mathcal{F}_l(X)}$. With the $\widehat{\mathcal{F}_l(X)}$, the block augments the fused features, which improves the robustness of the model.

formulated as:

$$f_{horizontal}\left(X_{i,j,c}\right) = \sum_{c=0}^{C} X_{i,j+o_{horizontal}(X_{i,j,c}),c} \cdot W_{c,:}^{horizontal} + b_{horizontal} \tag{5}$$

These features are fused to enhance the representation of input features as:

$$f_{global}(X) = f_{channel}(X) + f_{vertical}(X) + f_{horizontal}(X) \tag{6}$$

It can be noted that the computation complexity of the above operations only requires $O(HWC^2)$, which has a linear relation with the spatial dimension. Since the size of the channel dimension is obviously smaller than that of the spatial dimension, the computational complexity of the block is apparently smaller than the $O((HW)^2C)$ which other MLP-based models and Transformer-based models. This indicates the efficiency of our model is better than that of other MLP-based models and Transformer-based models.

Furthermore, to improve the accuracy of the feature with the global context, we utilize the boundary of the target to supervise the training. The process is formulated as:

$$f_{boundary}(X) = f_{global}(X) \cdot W_{boundary} \tag{7}$$

The $W_{boundary} \in \mathbb{R}^{d\times 1}$ denotes the weight of the Boundary MLP. It maps the feature with the global context to the boundary of the target. We use the boundary label of the target to supervise the prediction of the Boundary MLP. The boundary label is obtained by using Canny [30] operator on the ground truth. Note that the operation is only performed on the ground truth. Therefore, the obtained boundary only includes the boundary of the segmented target. Due to the import of extra supervised information, the SIE block improves the accuracy of the structural information. As the supervision process is only run in the training phase, it does not add extra computational costs in the inference phase.

3.3. Frequency spectrum fusion block

To preserve the detailed features of the image, the ISS models commonly fuse the features from the encoder and decoder with the skip connection. Unfortunately, the process imports the noise to the fused features. To reduce the noise of the skip connection, we develop the FSF block, which is used to remove the noise of the fused features in the frequency domain.

As shown in Fig. 3, the FSF block initially fuses the features from the encoder and SIE block. To reduce the noise of the fused feature, we use the parameterized matrix to adaptively compress the noise with element-wise multiplication, as illustrated in Fig. 5. In detail, given the fused features $X \in \mathbb{R}^{H\times W\times C}$, it is converted to the frequency spectrum along the spatial dimension by the 2D Fast Fourier Transform (FFT) [31]:

$$\mathcal{F}\left(X_{i,j,c}\right)_{u,v,c} = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} X_{i,j,c} e^{-z2\pi\left(\frac{i}{H}u+\frac{j}{w}v\right)} \tag{8}$$

where $0 \le i \le H-1$, $0 \le j \le W-1$, and $z^2 = -1$. After obtaining the frequency spectrum $\mathcal{F}\left(X_{i,j,c}\right) \in \mathbb{R}^{U\times V\times C}$ of feature X, we utilize the parameterized matrix $\omega \in \mathbb{R}^{U\times V\times C}$ to multiply the frequency spectrum:

$$\hat{\mathcal{F}} = \mathcal{F}(X) \odot \omega \tag{9}$$

where $\odot$ denotes the element-wise multiplication. Since the matrix ω covers all frequencies of the feature, the block can capture the rich context to reduce the noise of the fused feature.

Furthermore, to enhance the robustness of our model, we also utilize statistics of the frequency spectrum $\mathcal{F}(X)$ to model the distribution for the low-frequency spectrum. As shown in Fig. 2(a), the noise is mainly similar to the target in color and texture. Since the color and texture of the images are mainly affected by the low-frequency component, the varied low-frequency spectrum in the training phase can improve the robustness of the model. In this block, we resample the new statistics of the frequency spectrum to generate the new low-frequency spectrum in the training phase.

To model the distribution, we initially obtain the low-frequency spectrum $\mathcal{F}_l(X) \in \mathbb{R}^{U_l\times V_l\times C}$ with the low-frequency filter M as:

$$\mathcal{F}_l(X) = \mathcal{F}(X) \odot M \tag{10}$$

We compute the mean $\mu\left(\mathcal{F}_l(X)\right)$ and standard deviation $\sigma\left(\mathcal{F}_l(X)\right)$ of low-frequency spectrum $\mathcal{F}_l(X)$, which is utilized to obtain the corresponding uncertainty estimation. It is formulated as:

$$\mu\left(\mathcal{F}_l(X)\right) = \frac{1}{U_l V_l}\sum_{u=1}^{U_l}\sum_{v=1}^{V_l}\mathcal{F}_l(X) \tag{11}$$

$$\sigma(\mathcal{F}_l(X)) = \frac{1}{U_l V_l}\sum_{u=1}^{U_l}\sum_{v=1}^{V_l}[\mathcal{F}_l(X) - \mu(\mathcal{F}_l(X))]^2 \tag{12}$$

To resample a new low-frequency spectrum, we obtain new statistics with the uncertainty estimation $\sum_\mu \mathcal{F}_l(X)$, $\sum_\sigma \mathcal{F}_l(X)$ of $\mu\left(\mathcal{F}_l(X)\right)$ and $\sigma\left(\mathcal{F}_l(X)\right)$ respectively:

$$\Sigma_\mu^2(x) = \frac{1}{B}\sum_{i=1}^{B}[\mu\left(\mathcal{F}_l(X)\right) - \mathbb{E}[\mu(\mathcal{F}_l(X))]] \tag{13}$$

$$\Sigma_\sigma^2(x) = \frac{1}{B}\sum_{i=1}^{B}[\sigma\left(\mathcal{F}_l(X)\right) - \mathbb{E}[\sigma(\mathcal{F}_l(X))]] \tag{14}$$

The value of the uncertainty estimation denotes the possibility that the corresponding statistic changes. It is equivalent to data enhancement of the training samples, which improves the robustness of the model. To model the uncertainty, we randomly sample from the new feature statistics $\hat{\mu}\left(\mathcal{F}_l(X)\right)$ and $\hat{\sigma}\left(\mathcal{F}_l(X)\right)$. It is formulated as:

$$\hat{\mu}\left(\mathcal{F}_l(X)\right) = \mu\left(\mathcal{F}_l(X)\right) + \epsilon_\mu \Sigma_\mu \mathcal{F}_l(X), \quad \epsilon_\mu \sim \mathcal{N}(0,1) \tag{15}$$

$$\hat{\sigma}\left(\mathcal{F}_l(X)\right) = \sigma\left(\mathcal{F}_l(X)\right) + \epsilon_\sigma \Sigma_\sigma \mathcal{F}_l(X), \quad \epsilon_\sigma \sim \mathcal{N}(0,1) \tag{16}$$

The ϵ_μ and ϵ_σ both follow the standard Gaussian distribution. With the random coefficient, our model generates new feature statistics. With the new feature statistics, the block generates the new low-frequency spectrum $\widehat{\mathcal{F}_l(X)}$:

$$\widehat{\mathcal{F}_l(X)} = \hat{\mu}\left(\mathcal{F}_l(X)\right)\left(\frac{\mathcal{F}_l(X) - \mu\left(\mathcal{F}_l(X)\right)}{\sigma\left(\mathcal{F}_l(X)\right)}\right) + \hat{\sigma}\left(\mathcal{F}_l(X)\right) \tag{17}$$

The new low-frequency spectrum is fused with the high-frequency spectrum to generate the new frequency spectrum $\widehat{\mathcal{F}_l(X)}$. Finally, we utilize the 2D Inverse Fast Fourier Transform (IFFT) to convert the $\widehat{\mathcal{F}_l(X)}$ to the spatial domain, which generates new feature $\hat{X}$. Note that the process is only applied at the training phase, it gains the robustness of the model.

3.4. Loss function

In the ISS, cross-entropy is a general loss function to supervise the training of the ISS model [32]. However, in the low-contrast image, the segmented target is tiny, which leads to overfitting. To solve this problem, we jointly utilize the Dice loss function [33] and cross-entropy loss function [32] to train LCSeg-Net. Meanwhile, to make the structural information from the SIE block accurate, we use the boundary loss function to supervise all SIE blocks in the training phase.

Concretely, the hybrid loss L_{hybrid} is defined as follows:

$$L_{hybrid} = L_{Cross-entropy} + L_{Dice} + L_{Boundary} \tag{18}$$

where the $L_{Cross-entropy}$, L_{Dice}, and $L_{Boundary}$ represents the cross-entropy loss, soft dice-coefficient loss, and boundary loss respectively. The cross-entropy loss can be written as formula (19):

$$L_{cross-Entropy} = -\frac{1}{N}\sum_{c=0}^{C}\sum_{n=0}^{N} g_{n,c}^{original} \cdot log(o_{n,c}) \tag{19}$$

where the C and N denotes the number of the class and pixel respectively. Meanwhile, the $g_{n,c}^{original}$ and $o_{n,c}$ represents the ground truth of the input and segmentation results of LCSeg-Net respectively. The Dice loss L_{Dice} is formulated as follows:

$$L_{Dice} = 1 - \sum_{c=0}^{C}\frac{2(\sum_{n=0}^{N}(g_{n,c}^{original} \cdot o_{n,c}))}{\sum_{n=0}^{N}(g_{n,c}^{original} + o_{n,c})} \tag{20}$$

In the training process, the Boundary loss $L_{Boundary}$ is used to jointly optimize SIE blocks. It is shown in formula (21):

$$L_{Boundary} = \sum_{i=0}^{d}(1 - \sum_{c=0}^{C}\frac{2(\sum_{n=o}^{N}(g_{n.c}^{boundary} \cdot a_{n,c}^{i}))}{\sum_{n=o}^{N}(g_{n.c}^{boundary} + a_{n,c}^{i})}) \tag{21}$$

where the $g_{n,c}^{boundary}$ and $o_{n,c}^{i}$ represents the boundary label of the ground truth and the output of the ith block respectively. The boundary label of the ground truth is generated by the ground truth with the Canny operator [30]. Through the supervision of the hybrid loss function, our model predicts accurate segmentation results.

4. Experiments and results

4.1. Datasets

To fairly evaluate the proposed model, we use five public ISS datasets of the low-contrast image, CVC-ClinicDB [6], 2018 Data Science Bowl [34], International Skin Imaging Collaboration-2018 challenge [35], Magnetic Tile Defects Dataset [36], and Synthetic Smoke Datasets [8]. These datasets contain a lot of low-contrast segmentation objects, such as the polyp of the colon, and surface defect of magnetic. In addition, samples from these datasets were captured at different scales, lighting, and shooting angles, thus these datasets are very challenging. A brief introduction of them is as follows:

CVC-ClinicDB. The dataset is a standard benchmark of polyp segmentation. Especially, the polyp is one of the cancer precursors in the colon and rectum, hence accurate polyp segmentation will be helpful to reduce the emergence of colorectal cancer. To accurately segment the polyp, the dataset contains 612 images from 31 colonoscopy sequences, which are generated from 23 videos with standard white light colonoscopy interventions.

2018 Data Science Bowl. The dataset includes 670 nuclear images taken from microscopy. To effectively analyze the biological image, the nuclei segmentation of cells is an important step. Due to the difference in the morphology of cells, nuclei segmentation is a challenging task.

International Skin Imaging Collaboration (ISIC)-2018 challenge. The dataset is a public dermatology image analysis benchmark. To help improve the diagnostic accuracy of skin cancer and reduce melanoma deaths, the public dataset has 2596 different resolution images, each image contains only one lesion, and images are obtained by multiple types of dermoscopy.

Magnetic Tile Defects Dataset (MTDD). The magnetic tile is a key component of engines, while the magnetic tile has surface defects in the production process sometimes. To cover the defects of magnetic tile in the real assembly line, the dataset includes images with 1344 images. And the defects in these images have five types: blowhole, rack, uneven, break, and fray.

Synthetic Smoke Datasets (SSD). Smoke detection is critical for the safety of daily life. It ensures the monitoring system provides timely fire alarms in case of emergency. The dataset uses 20k pure smoke images with RGBA channels and a lot of background images with RGB channels to produce the dataset. To make the generated smoke resemble real transparent smoke, the smoke image is generated with the linear color composition of a pure smoke image and a background image. Meanwhile, to make the generated results more diverse, the linear color composition is performed on the three RGB channels respectively. The dataset includes about 70k images to train the model, and 3k images to validate and test the model.

4.2. Baselines

To verify the effectiveness of the proposed model, we choose some SOTA models to compare in the low-contrast image datasets. These models are listed as follows:

U-Net (MCCAI 2015) [10]**.** The U-Net is a popular ISS model that preserves detailed features with the skip connection.

Table 1
Statistics of datasets in CVC-ClincDB, 2018 Data Science Bowl, ISIC-2018 challenge, MTDD and SSD.

Dataset	Input size	No. of train	No. of valid	No. of test	No. of total
CVC-ClinicDB [6]	384 × 288	490	61	61	612
2018 Data Science Bowl [34]	256 × 256	536	67	67	670
ISIC-2018 challenge [35]	384 × 512	2078	259	259	2596
MTDD [36]	384 × 512	1075	133	133	1377
SSD [8]	256 × 256	70 632	1000	2000	73 632

UNet++ (TMI 2020) [25]**.** The UNet++ is the improvement of U-Net, which captures the multi-scale detailed feature with the dense skip connection. Meanwhile, the pruning scheme is used to accelerate the inference speed of the model.

Attention U-Net (MIDL 2018) [37]**.** The Attention U-Net focuses on the segmented objects of varying shapes and sizes by integrating the novel attention gate into the skip connection.

CE-Net (TMI 2019) [38]**.** The CE-Net captures more contextual information by incorporating a context encoder into the encoder–decoder network, which achieves comprehensive results on some ISS datasets.

KiU-Net (TMI 2022) [27]**.** The KiU-Net preserves the detailed features in deep layers with the overcomplete convolutional architecture, which projects the input image into a higher dimension.

MedFormer (ariXiv 2023) [29]**.** The MedFormer segments the objects accurately in diverse image datasets with the transformer structure block that incorporates desirable inductive bias, hierarchical modeling with linear-complexity attention mechanism, and multi-scale feature fusion.

4.3. Evaluation metrics

We use common metrics: Precision, Recall, Accuracy, Dice, and IoU to evaluate the performance of models that are widely used in ISS. IoU and Dice coefficient are used to measure the overlap between predicted and ground truth masks. As Accuracy is a commonly used evaluation parameter in experimental analysis. It is calculated as the ratio of the number of correct predictions to the total number of predictions made. IoU is particularly useful for semantic segmentation because it provides an overall measure of the model's ability to correctly identify and localize objects. It is calculated by dividing the intersection area of the two masks by their union area, which measures the extent of overlap between the predicted segmentation mask and the ground truth mask. Dice also known as the F1 score, assesses the similarity between the predicted and ground truth segmentation masks. It is calculated by comparing the size of the intersection of the masks with the sum of the sizes of the individual masks. The Dice coefficient is valuable in semantic segmentation because it quantifies the agreement between the predicted and ground truth masks, considering both positive and negative predictions. It is particularly useful when dealing with imbalance issues. The Dice coefficient effectively captures the performance of the segmentation model in terms of true positives and false negatives, providing a balanced evaluation metric. Accuracy, Dice, and IoU range from 0 to 1, with 1 indicating a perfect match and 0 completely different. Higher values indicate better accuracy. Specially, due to the sensitivity of Dice for small differences, it is widely applied in medical image analysis and other segmentation tasks. The definitions of these metrics are shown as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{22}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{23}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{24}$$

$$\text{Dice} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{25}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{26}$$

where TP, FP, and FN denote the number of correctly detected pixels, falsely detected pixels, and undetected pixels respectively [1]. To evaluate the efficiency of the model, we select the Parameters (Params), Frame Per Second (FPS) and Floating Point Operations (FLOPs) of ISS models as the metric.

4.4. Setup and details

All the experiments are carried out on NVIDIA RTX 3090 GPU with 24 GB memory and Ubuntu Server 16.04 LTS system using the framework PyTorch 1.7.1. Due to the variable size of samples from the ISIC-2018 challenge and Magnetic Tile Defects Datasets, we resize images from the ISIC-2018 challenge and MTDD to 384 × 512. Meanwhile, we use 80% of the images for training, 10% for validation, and 10% for testing in all the datasets except SSD. In the original SSD, there are three testing sets, DS1, DS2, and DS3. Therefore, we choose DS1 as the validation set. The DS2 and DS3 are set as testing datasets. To avoid overfitting, we apply the usual data augmentation techniques to the training dataset. The data augmentation techniques include center crop, random rotation, transpose, and Gaussian noise. We set the batch size to 32 and the learning rate to 0.001 for the 2018 Data Science Bowl and SSD, the batch size to 8, and the learning rate to 0.0001 for other datasets by the reason of the bigger size of images of other datasets. All experiments are trained in 50 epochs with the early stopping and Adam optimizer. In these experiments, all models except KiU-Net are trained with the same configuration. As KiU-Net has an extremely large memory requirement, it is very difficult to achieve, as a result, the configurations for KiU-Net are the same as that of the original paper. The statistics of the datasets are shown in Table 1.

4.5. Experiments on CVC-ClinicDB dataset

CVC-ClinicDB is a public polyp segmentation dataset that collects polyps from the colonoscopy. In this experiment, we compare the results of the proposed LCSeg-Net with that of SOTA ISS models. As shown in Table 2, LCSeg-Net outperforms all the SOTA models on this dataset in all metrics. In the comparison, the performance of the proposed model has a large improvement over that of U-Net, U-Net++, and KiU-Net. These models only concentrate on preserving detailed features and ignore the noise of fused features in their skip connections. As a result, the Accuracy, Precision, IoU, and Dice of these models are lower than that of LCSeg-Net. Our model effectively reduces the impact of the noise with the FSF block. This is because it fuses the feature with the parameterized matrix, which suppresses the noise and emphasizes useful information. In addition, Attention U-Net and CE-Net further improve the ISS performance with the additive attention gate and atrous convolution. However, these models segment the low-contrast polyp only with the local features. The indistinguishable boundary of the polyp reduces the performance of these models. In contrast, LCSeg-Net utilizes the global context to enhance the structural information, which makes the performance of the LCSeg-Net higher than that of Attention U-Net and CE-Net. Compared with MedFomer, our model yields better performance in all metrics, achieving Dice and Accuracy as high as 95.82% and 99.25% respectively. This is because the MedFormer loses useful information in order to reduce the computational

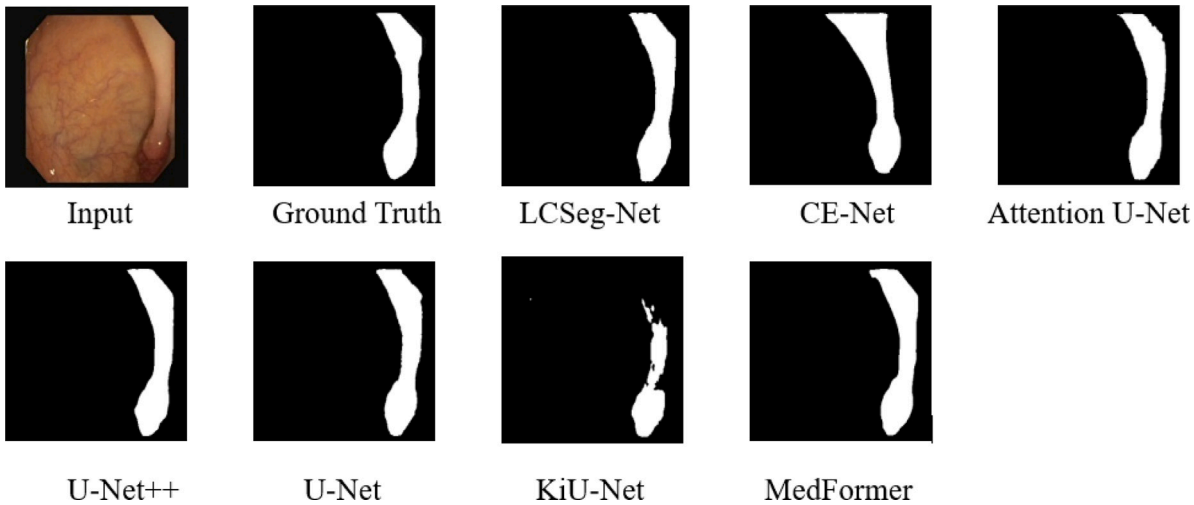

Fig. 6. Sample results of polyp segmentation. From left to right: input image, ground truth, SOTA results obtained by LCSeg-Net, CE-Net, Attention U-Net, U-Net++, U-Net, KiU-Net, MedFormer.

Table 2
Performance of polyp segmentation. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
U-Net [10]	ResNet 34	0.9806	0.8726	0.8095	0.8930	0.8827
UNet++ [25]	ResNet 34	0.9817	0.8763	0.8115	0.8597	0.9192
CE-Net [38]	ResNet 34	0.9872	0.9317	0.8727	0.9165	0.9484
Attention-Unet [37]	–	0.9836	0.8954	0.8358	0.9016	0.9046
KiU-Net [27]	–	0.9785	0.8571	0.7677	0.8640	0.9216
MedFormer [29]	–	0.9901	0.9434	0.8950	0.9423	0.9461
LCSeg-Net	ResNet 34	**0.9925**	**0.9582**	**0.9202**	**0.9585**	**0.9587**

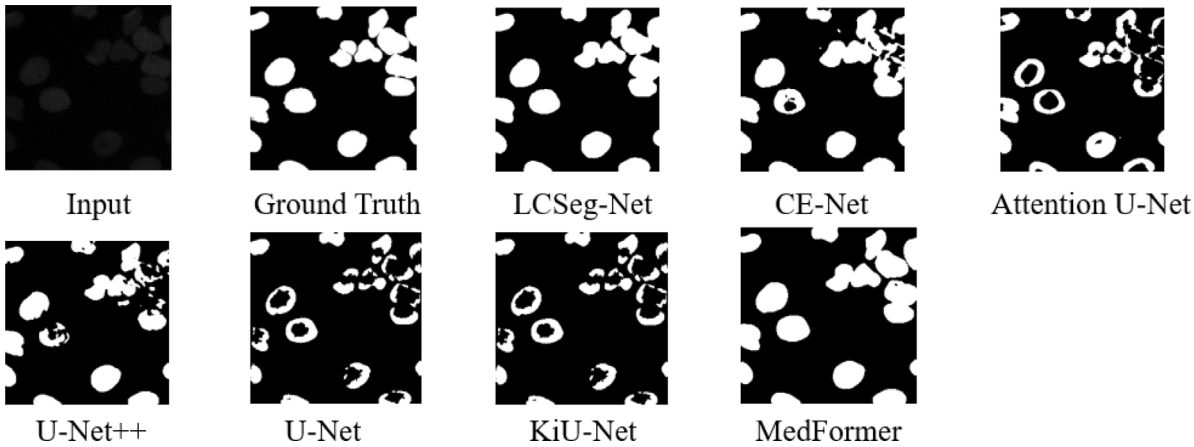

Fig. 7. Sample results of nuclei segmentation. From left to right: input image, ground truth, SOTA results obtained by LCSeg-Net, CE-Net, Attention U-Net, U-Net++, U-Net, KiU-Net, MedFormer.

cost with the B-MSA block. Differently, the proposed extracts the global contexts without changing the resolution of the feature, which accurately predicts the polyp. Furthermore, from Fig. 6, we observe LCSeg-Net accurately segments the polyp even if the boundary of the object is blurry.

4.6. Experiments on 2018 Data Science Bowl dataset

The results in Table 3 show that the performance of the proposed model outperforms other models in all metrics except for the Recall which is only second to that of KiU-Net. This is due to the over-complete convolutions of the KiU-Net, it saves a lot of detailed features. With these detailed features, the KiU-Net improves the Recall. However, the over-complete convolutions also introduce a lot of noise into the fused features. Meanwhile, the upsamle operation of the over-complete convolution enlarges the impact of the noise. Thus the KiU-Net achieves poor performance of Precision with 88.04%. Differently, our model reduces the noise with the SIE block, which accurately distinguishes the noise and nuclei. This makes the performance of LCSeg-Net in Accuracy, Dice, IoU, and Precision outperforms KiU-Net by a large margin with 0.27% 7.82%, 8.59%, and 4.96% respectively. Among all the methods in comparison, MedFormer and CE-Net achieve second-best and third-best performance in Dice and IoU respectively, which proves increasing the size of the ERF improves the performance

Table 3
Performance of nucleus segmentation. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
U-Net [10]	ResNet 34	0.9736	0.8875	0.8080	0.9208	0.8722
UNet++ [25]	ResNet 34	0.9726	0.8868	0.8141	0.9188	0.8740
CE-Net [38]	ResNet 34	0.9748	0.9106	0.8365	0.9359	0.8871
Attention-Unet [37]	–	0.9750	0.8879	0.8163	0.9181	0.8705
KiU-Net [27]	–	0.9786	0.8394	0.7648	**0.9702**	0.8804
MedFormer [29]	–	0.9812	0.9164	0.8493	0.9361	0.9265
LCSeg-Net	ResNet 34	**0.9813**	**0.9176**	**0.8507**	0.9466	**0.9300**

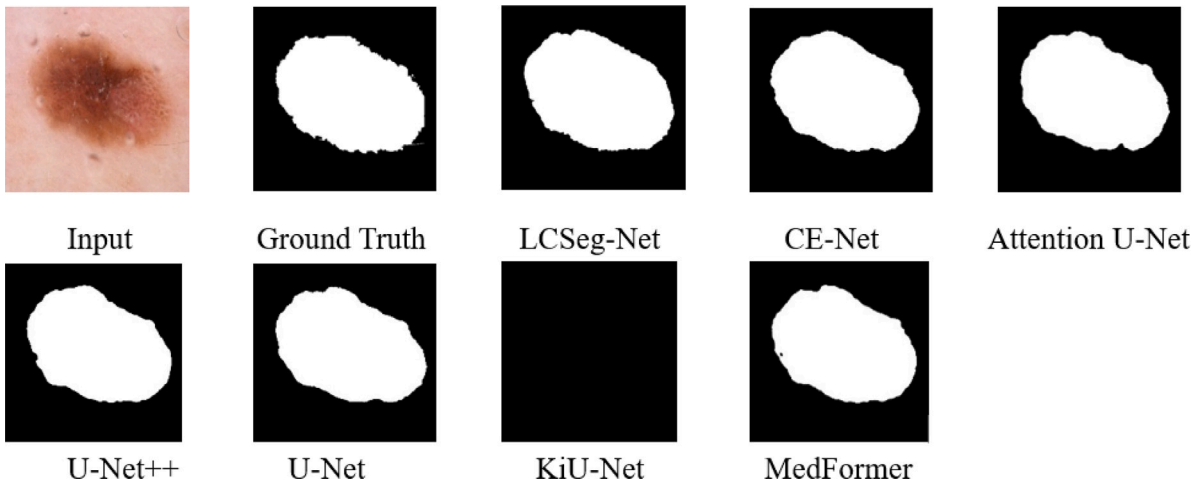

Fig. 8. Sample results of skin lesion segmentation. From left to right: input image, ground truth, SOTA results obtained by LCSeg-Net, CE-Net, Attention U-Net, U-Net++, U-Net, KiU-Net, MedFormer.

Table 4
Performance of skin cancer segmentation. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
U-Net [10]	ResNet 34	0.9484	0.8746	0.8025	0.9066	0.8837
UNet++ [25]	ResNet 34	0.9517	0.8834	0.8122	0.9061	0.8997
CE-Net [38]	ResNet 34	0.9678	0.9016	0.8243	0.8795	0.9303
Attention-Unet [37]	–	0.9529	0.8834	0.8149	0.8901	0.9153
KiU-Net [27]	–	0.8416	0.7017	0.5647	0.8301	0.6813
MedFormer [29]	–	0.9667	0.8982	0.8195	0.9045	0.9136
LCSeg-Net	ResNet 34	**0.9901**	**0.9262**	**0.88530**	**0.9095**	**0.9436**

of the ISS model in the dataset. While the best Accuracy, Dice and IoU of LCSeg-Net prove it accurately segments the low-contrast target with the global context. Through the analysis of experimental results, we find LCSeg-Net effectively segments nuclei with the global context. The qualitative results of 2018 Data Science Bowl show that the proposed model produces high-quality masks for nuclei in Fig. 7.

4.7. Experiments on ISIC-2018 challenge dataset

As shown in Table 4, we compare our model with 6 SOTA models on the dataset ISIC-2018 challenge. In the dataset, skin lesions have ambiguous contours, which reduces the accuracy of the ISS model with the local feature. In the comparison, U-Net, U-Net++, Attention U-Net, and KiU-Net only extract the local feature with the traditional convolution, which reduces the accuracy of segmenting skin lesions. Moreover, CE-Net and MedFormer improve the performance by enlarging the ERF. Compared with models based on the traditional convolution, which only extracts the local feature, we find CE-Net and MedFormer achieve better Accuracy, Dice and IoU. These metrics represent the comprehensive performance of the ISS model. It proves that the increment of the ERF effectively improves the performance in segmenting the skin lesion. Furthermore, MedFormer is slightly smaller than CE-Net in Accuracy, Dice, IoU, and Precision. To reduce the computational cost, MedFormer compresses the spatial size of the feature map. This results in the performance of MedFormer being slightly less than that of CE-Net in Accuracy, Dice, IoU, and Precision. In contrast, LCSeg-Net extracts the global features with the original spatial size of the feature, which achieves the best score in all metrics. From the observation in Fig. 8, even though the contour of the skin lesion is blurry, the proposed model still accurately segments objects with the global context.

4.8. Experiments on Magnetic Tile Defects dataset

Magnetic Tile Defects Dataset aims to develop an automatic algorithm for the detection of surface defects of magnetic tiles, which reduces labor costs. The dataset contains five types of defects, some of which are similar to the magnetic tile. Therefore, due to complex morphologies of defects and undistinguishable defects, it is hard to accurately segment the defect in the magnetic tile. We compare the results of the proposed model with that of 6 SOTA defect detection models on the Magnetic Tile Defects Dataset, the results are shown in Table 5. The LCSeg-Net achieves SOTA results and outperforms other SOTA models on almost all metrics by a large margin. Compared with U-Net, U-Net++, CE-Net, Attention U-Net, CE-Net, and KiU-Net, the proposed model achieves a large improvement in the performance. In the comparison, U-Net, U-Net++, and CE-Net poorly perform on the

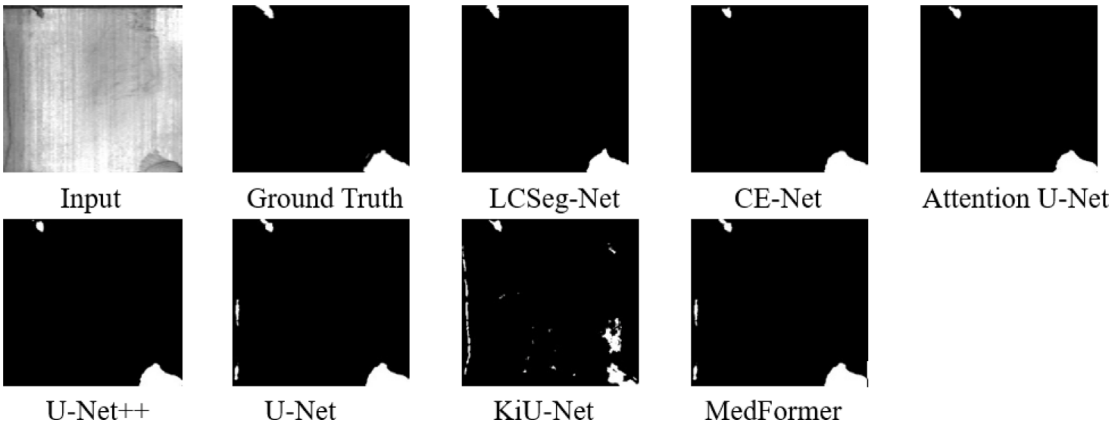

Fig. 9. Sample results of magnetic tile segmentation. From left to right: input image, ground truth, SOTA results obtained by LCSeg-Net, CE-Net, Attention U-Net, U-Net++, U-Net, KiU-Net, MedFormer.

Table 5
Performance of surface defect segmentation of magnetic tile. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
U-Net [10]	ResNet 34	0.9513	0.6453	0.4417	0.6324	0.5842
UNet++ [25]	ResNet 34	0.9705	0.7671	0.6778	0.9145	0.8216
CE-Net [38]	ResNet 34	0.9764	0.7512	0.6608	0.9195	0.8014
Attention-Unet [37]	–	0.9786	0.7922	0.7095	**0.9458**	0.8478
KiU-Net [27]	–	0.9442	0.6241	0.4689	0.6970	0.6182
MedFormer [29]	–	0.9862	0.9047	0.8293	0.9043	0.9367
LCSeg-Net	ResNet 34	**0.9906**	**0.9316**	**0.8734**	0.9341	**0.9547**

Table 6
Performance of smoke semantic segmentation. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
U-Net [10]	ResNet 34	0.8774	0.8017	0.6801	0.8284	0.8005
UNet++ [25]	ResNet 34	0.8867	0.8036	0.6831	0.8253	0.7885
CE-Net [38]	ResNet 34	0.8803	0.7945	0.6729	0.8269	0.7913
Attention-Unet [37]	–	0.8963	0.8129	0.6961	0.8247	0.8237
KiU-Net [27]	–	0.7506	0.6112	0.4654	0.7992	0.5823
MedFormer [29]	–	0.8220	0.7281	0.5730	0.7299	0.7293
LCSeg-Net	ResNet 34	**0.9073**	**0.8385**	**0.7228**	**0.8347**	**0.8544**

dataset, which is because these models lack structural information with the local feature. Furthermore, the overcomplete convolution of KiU-Net only focuses on preserving the detailed feature, while it generates a lot of noise. Therefore, on the dataset, KiU-Net only achieves 0.6241 in Dice. In addition, even though the Recall of Attention U-Net is slightly higher than that of our model, it still has the second performance in Recall, and an improvement of 1.20% in Accuracy, 13.94% in Dice, 16.39% in IoU, and 10.69% in Precision over that of Attention U-Net. The higher Recall means Attention U-Net detects more surface defects, and yet the lower Precision means the model also recognizes more noisy pixels as object pixels. Meanwhile, higher scores in Accuracy, Dice, and IoU mean LCSeg-Net has better comprehensive performance. Different from MedFormer which compresses the spatial size of the feature with the B-MSA block, LCSeg-Net keeps the spatial size of the feature with the SIE block. Therefore, it achieves better performance in all metrics over MedFormer. From Fig. 9, we also find LCSeg-Net accurately segments defects of magnetic tiles.

4.9. Experiments on synthetic smoke datasets

Smoke semantic segmentation is an indispensable visual task in our daily life, the reliable smoke semantic segmentation models reduce the risk of fire disasters [39]. In the dataset, smoke is translucent which makes the textures of smoke mix with that of the background. As a result, the appearance of the smoke is easily affected by the complex environments. It makes the ISS models need to reduce the effect of the background. As shown in Table 6, Attention U-Net achieves the second-best performance in Accuracy, Dice, IoU, and Precision. It proves that adding extra context to the fused feature in the skip connection effectively improves the performance of the model on the dataset. However, the performance of the Attention U-Net is lower than that of LCSeg-Net. Especially, the better Precision of LCSeg-Net proves it removes more noise of the fused feature in the skip connection. Meanwhile, LCSeg-Net utilizes structural information to enhance the representation of the features, which improves the performance of the model. As a result, the proposed model achieves the best performance in all metrics. From Fig. 10, we observe that LCSeg-Net effectively segments the smoke and avoids the influence of the noise, which makes it more reliable in the fire alarm and smoke alarm system. As shown in Fig. 10, most of models generate poor results except LCSeg-Net and U-Net++. In detail, our LCSeg-Net obtains more accurate boundary of the target. Especially for the right of the smoke, which has higher transparency, is difficult to be accurately segmented. LCSeg-Net accrately capture the shape of the smoke and details of the boundary than U-Net++ does.

4.10. Experiments on sequential images of smoke videos

Due to the gap between the synthetic and the real smoke data, we compare the proposed model with other SOTA models. The data is collected from Internet. We resize these images to the same size

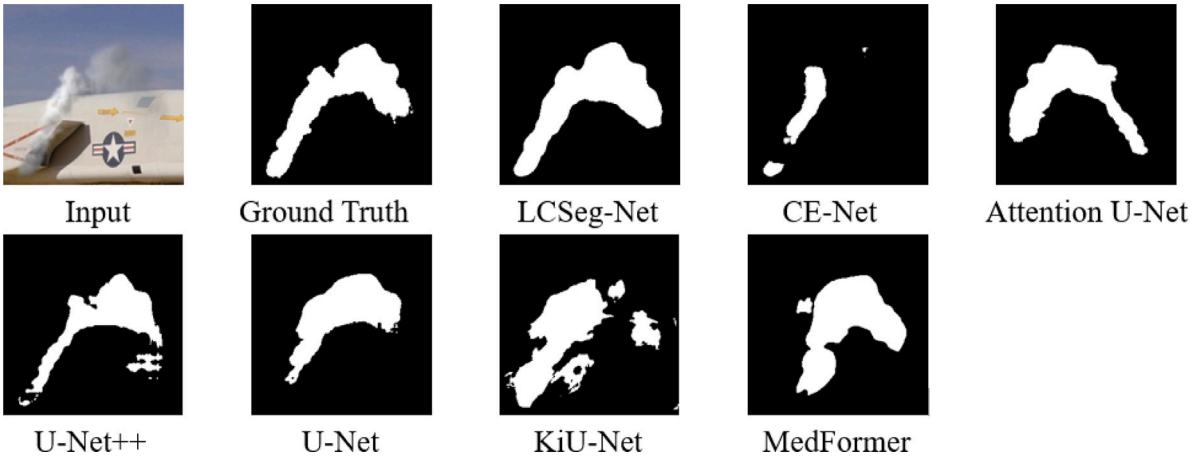

Fig. 10. Sample results of smoke semantic segmentation. From left to right: input image, ground truth, SOTA results obtained by LCSeg-Net, CE-Net, Attention U-Net, U-Net++, U-Net, KiU-Net, MedFormer.

Fig. 11. Test results on realistic smoke videos. From top to bottom: input image, results obtained by LCSeg-Net, Attention U-Net, CE-Net, KiU-Net, U-Net, UNet++, and MedFormer.

Table 7
FPS of smoke semantic segmentation. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	FPS (↑)
U-Net [10]	ResNet 34	144.28
UNet++ [25]	ResNet 34	126.00
CE-Net [38]	ResNet 34	124.66
Attention-Unet [37]	–	106.62
KiU-Net [27]	–	17.17
MedFormer [29]	–	49.44
LCSeg-Net	ResNet 34	71.72

as the synthetic data. Since these real images have no ground truth, we only conduct visual comparisons. Specifically, we utilize the above models, which are trained on synthetic data, to segment real smoke. Fig. 11 shows the segmented results by seven models. The first row shows five smoke frames from two videos. The other rows show the predictions of smoke segmentation by seven models. Meanwhile, the first two columns and the last three columns show the segmentation results from the black and white smoke video respectively. It can be noted that due to the disturbance of window refraction in the first two columns, the texture of the black smoke is broken. In addition, the contrast between the white smoke and the bright background is low. Hence, the segmentation of the real smoke is difficult.

Through observation of segmentation results, we find LCSeg-Net accurately segments the black smoke. As shown in the first two columns of the figure, KiU-Net, U-Net, UNet++ and MedFormer hardly segment smoke. In addition, our LCSeg-Net captures more accurate boundary shape than other models do. This is because the model can capture accurate structural features. As shown in the other columns of the Fig. 11, the accuracy of results from all models has significantly decreased except for our LCSeg-Net. It indicates that segmenting white smoke is challenging, while LCSeg-Net accurate recognize the region of the smoke. This advantage stems from the robustness of our model to the variations of texture and color.

Furthermore, the test time is important for smoke detection. Therefore, we compare the Frame Per Second (FPS) of the above models. As shown in Table 7, the FPS of LCSeg-Net is lower than that of UNet, UNet++, Attention U-Net, CE-Net. This is because LCSeg-Net needs extra operations that extract global contexts, fuse multi-scale features, and transform features to the frequency domain. Even though the FPS 71.72 of our model is lower than that of UNet, UNet++, Attention U-Net, CE-Net, it is sufficient for the practical application [40]. Meanwhile, the detection effect of LCSeg-Net is much better than that of other models in real scenarios. As a result, compared with other models, we believe that LCSeg-Net is more suitable for the smoke segmentation.

4.11. Results and analysis

To evaluate the effectiveness and generality of the proposed model, we compare our model with 6 SOTA models on the five public datasets: CVC-ClinicDB, 2018 Data Science Bowl, ISIC-2018 challenge, MTDD, and SSD. In the comparison, U-Net, U-Net++, Attention U-Net, and KiU-Net are the models that only with the local feature. Results show that our model performs much better than these models on the CVC-ClinicDB, 2018 Data Science Bowl, ISIC-2018 challenge, and MTDD. It is caused by the ambiguous boundary of the segmented object in these datasets. Meanwhile, the local feature ignores the structural information. Therefore, our model and the MedFormer perform well on these datasets with the global feature. However, the B-MSA block of MedFormer reduces the spatial size of the feature map, while the SIE block saves the spatial size of the feature map. As a result, the performance of the proposed model is better than that of MedFormer on these datasets. In addition, in the experiments of the Synthetic Smoke Dataset, due to the translucency of the smoke, the complex noise of the background reduces the performance of U-Net, U-Net++, KiU-Net, and MedFormer. This is due to the noise of the fused feature in the skip connection. To reduce the noise of the fused feature, Attention U-Net uses the convolution layer to generate the matrix to guide the feature fusion. Differently, our model guides the fusion with the parameterized matrix in the frequency domain. Since the matrix covers all the frequencies, it captures both long-term and short-term relationships, which provides more context in the skip connection than Attention U-Net does. As a result, compared with Attention U-Net, LCSeg-Net achieves better performance in all metrics.

4.12. Efficiency analysis on CVC-ClinicDB dataset

To evaluate the efficiency of the ISS models, we compare the FLOPs, Params, Dice and FPS of the proposed LCSeg-Net with that of other baseline models. As shown in Table 8, the LCSeg-Net achieves the best FLOPs of 12573.72 and Dice of 95.82%, which proves our model effectively and efficiently obtains accurate results of ISS in the low-contrast image. Meanwhile, LCSeg-Net obtains the second-best Params, while KiU-Net obtains the best Params. This is because KiU-Net mainly utilizes the upsampling operation to obtain the over-complete features, which saves a lot of parameters. However, the computation complexity of KiU-Net is higher than that of LCSeg-Net by 37 times due to numerous operations involving the over-complete features. At the same time, the Dice score of LCSeg-Net is higher than that of KiU-Net. It indicates our model obtains better comprehensive performance than KiU-Net does. Compared with the transformer-based MedFormer, the proposed model has better Dice with lower FLOPs and Params. This is because the self-attention mechanism of the transformer-based models needs a lot of computation costs to capture the global context. In contrast, our model obtains the global context with the SIE block, which has low computational costs. As shown in the Table, we find that the FPS of KiU-Net and MedFormer are much lower than that of the proposed LCSeg-Net. This is due to the high resolution of the over-complete features and computational complexity of KiU-Net and MedFormer respectively. In addition, the FPS of the U-Net and UNet++ achieve the best and second scores. This is because these models lack the extraction of contextual information, which significantly reduces the Dice. Though our proposed LCSeg-Net does not achieve the fastest running speed, it still has excellent performance on the metrics of Flops, Params. It denotes the LCSeg-Net have low computational cost. With the optimization of the computational for the proposed model in the future, there is still alot of room for improvement in term of running speed. In summary, the LCSeg-Net obtains accurate segmentation results with low computational costs and parameters, which demonstrates the model is an efficient and effective solution for ISS in the low-contrast image.

4.13. Ablation analysis on CVC-ClinicDB dataset

To verify the rationality of the overall architecture of LCSeg-Net and different blocks of the model to ISS in the low-contrast image, we conduct extensive ablation experiments on the CVC-ClinicDB dataset. In the ablation experiments, all models leverage ResNet 34 [17] as the encoder. We also provide different design choices and show their impact on the results.

Effects of the SIE block. To evaluate the effects of the SIE block and different parts of the block. We compare the LCSeg-Net with four variants of the proposed network by removing the SIE block, and different parts of the SIE block. As shown in Table 9, LCSeg-Net achieves higher metrics than Variant 4 does in Accuracy, Dice, IoU, Recall, and Precision. It indicates the SIE block effectively improves the accuracy of segmentation in the low-contrast image with enhanced structural information. Furthermore, we find the Dice of Variant 2 is lower than that of LCSeg-Net by 4.45%, which is only lower than that of Variant 4. The result suggests the Vertical MLP and the Horizontal MLP are the most important parts of the SIE block. This is because

Table 8
Comparison on CVC-ClinicDB. The comparison metrics are the Params, FLOPs, and Dice. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance. The ↓ represents that the lower the metric score, the better the model performance.

Method	Backbone	FLOPs (↓)	Params (↓)	Dice (↑)	FPS (↑)
U-Net [10]	ResNet 34	13246.48	24.44	0.8726	**61.93**
UNet++ [25]	ResNet 34	31 134.08	26.07	0.8763	60.45
CE-Net [38]	ResNet 34	15 020.02	29.00	0.9317	54.74
Attention-Unet [37]	–	112 441.24	34.89	0.8954	56.89
KiU-Net [27]	–	474 775.67	**0.21**	0.8571	10.55
MedFormer [29]	–	36 181.89	28.07	0.9434	11.30
LCSeg-Net	ResNet 34	**12 573.72**	22.60	**0.9582**	38.38

Table 9
Performance of ablation study on the SIE block. The best and second-best results in each metric are highlighted in bold and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
Variant 1 (-Channel MLP)	0.9865	0.9298	0.8993	0.9307	0.9295
Variant 2 (-Vertical MLP – Horizontal MLP)	0.9858	0.9137	0.8823	0.9134	0.9142
Variant 3 (-Boundary MLP)	0.9894	0.9495	0.9012	0.9489	0.9503
Variant 4 (-SIE block)	0.9843	0.9071	0.8791	0.9108	0.9053
LCSeg-Net	**0.9925**	**0.9582**	**0.9202**	**0.9585**	**0.9587**

Table 10
Performance of polyp segmentation. All models are supervised with same loss function. The best and second-best results in each metric are highlighted in **bold** and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Backbone	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
U-Net [10]	ResNet 34	0.9813	0.8847	0.7963	0.9049	0.8878
UNet++ [25]	ResNet 34	0.9827	0.8886	0.8063	0.9145	0.8933
CE-Net [38]	ResNet 34	0.9910	0.9359	0.8823	0.9488	0.9516
Attention-Unet [37]	–	0.9854	0.8993	0.8210	0.9315	0.8993
KiU-Net [27]	–	0.9794	0.8723	0.7795	0.8620	0.9139
MedFormer [29]	–	0.9909	0.9482	0.9022	0.9428	0.9545
LCSeg-Net	ResNet 34	**0.9925**	**0.9582**	**0.9202**	**0.9585**	**0.9587**

the enhancement of the structural information mainly relies on the context along the spatial dimension. The Vertical MLP and Horizontal MLP extract the global context to improve the performance of the LCSeg-Net. Furthermore, LCSeg-Net has improvements in the Dice with 0.0087 over that of Variant 3. It proves that the Boundary MLP further improves the accuracy of structural information with the supervision of the boundary, which helps the model obtain high-quality predictions. Meanwhile, since the Boundary MLP is inference-free, it is cost-free in the inference phase.

In addtion, to further illustrate the effect of the SIE block, we compare LCSeg-Net with other SOTA models. These models are trained with extra boundary information that is the same to that of LCSeg-Net. In detail, these models are supervised by the boundary information in each scale with the convolution. The results are shown in Table 10. As shown in the table, we find these models all achieve improvements in the performance with the supervision of the boundary information. However, the performance of these models still are lower than that of LCSeg-model. For example, even though U-Net obtains the same supervision as LCSeg-Net, the convolution of the model cannot effectively extract the global context. It makes the improvement of the model is limited. We also visualize the segmentation results of these models in Fig. 12. The input image is the same as the input in Fig. 6. As shown in Fig. 12, these models generate more accurate predictions than that of Fig. 6, especially for the boundary of the polyp. However, these models lack effective extraction measures of the structural information, which leads to the prediction of LCSeg-Net still better than that of these models.

Effects of the FSF block. To assess the effectiveness of the FSF block and the resampling strategy of the block, we conduct ablation experiments on the LCSeg-Net and two variants of the model by removing the FSF block, and resampling strategy of the block.

Table 11 displays the outcomes of our experiments where LCSeg-Net yields the best results. As shown in the table, the scores of Variant 2 are lower than that of LCSeg-Net in Accuracy, Dice, IoU. Due to the absence of the FSF block, Variant 2 directly fuses features from the encoder and the SIE block. It ignores the noise of the features from the encoder, which reduces the performance. Conversely, LCSeg-Net utilizes the FSF block to adaptively fuse features. The parameterized matrix of the FSF block suppresses the noise of the fused features in the frequency domain, which improves the accuracy of the segmentation results. Additionally, we find that LCSeg-Net achieves an improvement of 1.64% on Dice over Variant 1. It proves the resample training strategy improves the performance of the FSF block. This is because the noise is similar to the segmented target of the low-contrast image on texture. Since the texture of the image is mainly dependent on the low-frequency components of the feature, the resampling of the low-frequency spectrum effectively augments the texture of the training sample. As a result, the resampling strategy improves the robustness of LCSeg-Net, which obtains high-quality predictions.

The results of the ablation experiment demonstrate the proposed SIE block and FSF block are helpful for the ISS task in the low-contrast image. The SIE block is effective in enhancing the structure information and the FSF block is helpful to decrease the negative impact of noisy pixels that are like segmented targets. With the combination of these blocks into the basic model, we obtain the best Accuracy, Dice, IoU model LCSeg-Net on the CVC-ClinicDB dataset. It is worth mentioning that the Boundary MLP of the SIE block and the resampling strategy of the FSF block are inference-free, which keeps the efficiency of the LCSeg-Net.

5. Conclusions

In this work, we propose a model LCSeg-Net to improve the performance of ISS in low-contrast images. The model improves the accuracy of segmenting the low-contrast target that has ambiguous boundaries

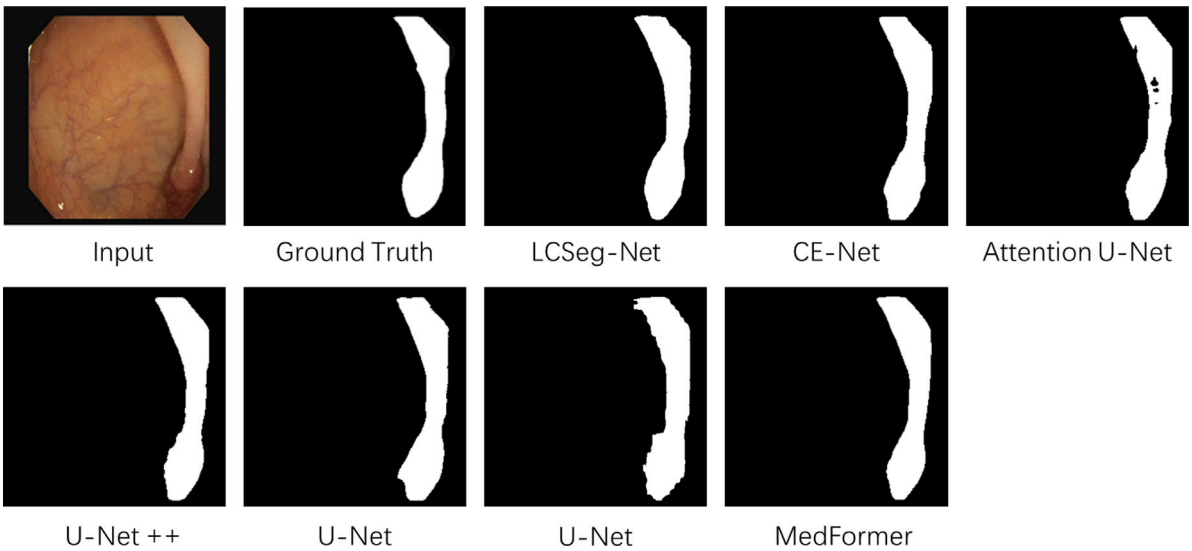

Fig. 12. Sample results of polyp segmentation. From left to right: input image, ground truth, SOTA results obtained by LCSeg-Net, CE-Net, Attention U-Net, U-Net++, U-Net, KiU-Net, MedFormer.

Table 11
Performance of ablation study on the FSF block. The best and second-best results in each metric are highlighted in bold and red. The ↑ represents that the bigger the metric score, the better the model performance.

Method	Accuracy (↑)	Dice (↑)	IoU (↑)	Recall (↑)	Precision (↑)
Variant 1 (Resampling stratege)	0.9918	0.9418	0.8913	0.9387	0.9437
Variant 2 (-FSF block)	0.9879	0.9302	0.8734	0.9430	0.9177
LCSeg-Net	**0.9925**	**0.9582**	**0.9202**	**0.9585**	**0.9587**

and indistinguishable noise. In this model, the global context is extracted to enhance the structural information, which accurately localizes the boundary of the low-contrast target. Meanwhile, due to the similar texture between the noise and the segmented target, to avoid segmenting the noise as the target, the proposed model suppresses the noise of the fused feature in the skip connection. Specially, in the model, we design the SIE block and FSF block, which are utilized to enhance the structural information and adaptively fuse the features in the frequency domain. We conduct extensive experiments on five public datasets, and the results demonstrate that the LCSeg-Net achieves the best comprehensive performance for low-contrast image segmentation. It indicates the proposed model has the potential to be applied in various domains such as medical diagnosis, industrial defect detection, and fire alarm. However, the proposed model is designed to process static images, while there is a significant amount of dynamic video data. Therefore, for future work, we plan to study how to utilize temporal features to enhance the performance of ISS in low-contrast images of videos. Meanwhile, due to the cost of labeling, we also plan to study how to reduce the dependence of the model on the labeled data.

CRediT authorship contribution statement

Haochen Yuan: Methodology, Software, Visualization, Writing – original draft, Conceptualization, Data curation, Formal analysis, Writing – review & editing. **Junjie Peng:** Conceptualization, Formal analysis, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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